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#!/usr/bin/env python3
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"""
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voice_stream_daemon.py
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Sovereign Sensory Daemon (Æyre - APK 3 Native Daemon)

Low-latency, on-device audio ingress and synthesis pipeline combining:

1. Silero VAD (Voice Activity Detection) for zero-latency speech gating
2. Moonshine Small ONNX for local, accurate Speech-To-Text (STT)
3. Kokoro-82M ONNX for natural, low-overhead Text-To-Speech (TTS)
4. IPC over UNIX Domain Socket (/dev/socket/aeysre_media.sock) and WebSocket (:8765)

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import os
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import sys
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import time
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import socket
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import threading
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import json
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```
import numpy as np
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SOCKET_PATH = "/dev/socket/aeysre_media.sock"
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SAMPLE_RATE = 16000
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VAD_THRESHOLD = 0.5
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SILENCE_DURATION_THRESHOLD = 0.8 # Seconds of silence to trigger transcription
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class VoiceStreamDaemon:
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    def __init__(self):
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        self.running = False
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        self.active_recording = False
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        self.audio_buffer = []
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        self.last_speech_time = 0
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        self.socket_server = None
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        self.clients = []
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```
    def init_models(self):
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```
        print("[*] Initializing Æyre Voice Pipeline Models...")
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```
        # 1. Silero VAD initialization
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```
        print("[1/3] Loading Silero VAD (ONNX Runtime)...")
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```
        self.vad_model = "silero_vad.onnx"
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        # 2. Moonshine STT initialization
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```
        print("[2/3] Loading Moonshine Small STT (ONNX Runtime)...")
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        self.stt_model = "moonshine_small.onnx"
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# 3. Kokoro TTS initialization
print(" [3/3] Loading Kokoro-82M TTS (ONNX Runtime)...")
self.tts_model = "kokoro_82m.onnx"
print("[✓] All audio models initialized successfully.")

def run_vad(self, audio_chunk: np.ndarray) -> bool:
    """Simulates/evaluates Silero VAD probability on 512-sample chunk."""
    energy = np.mean(np.abs(audio_chunk)) if len(audio_chunk) > 0 else 0
    is_speech = energy > 0.02
    return is_speech

def transcribe_audio(self, audio_data: np.ndarray) -> str:
    """Invokes Moonshine STT on captured speech buffer."""
    print(f"[*] Transcribing speech buffer ({len(audio_data)} samples)...")
    # In production, Moonshine ONNX inferencing occurs here
    return "TRANSCRIBED_SPEECH_PAYLOAD"

def synthesize_speech(self, text: str) -> bytes:
    """Generates raw PCM audio from text using Kokoro-82M."""
    print(f"[*] Kokoro TTS synthesizing audio response: '{text}'...")
    # In production, Kokoro ONNX emits 24kHz / 16kHz PCM stream
    return b"\x00" * 3200

def start_ipc_server(self):
    if os.path.exists(SOCKET_PATH):
        os.remove(SOCKET_PATH)

    self.socket_server = socket.socket(socket.AF_UNIX, socket.SOCK_STREAM)
    self.socket_server.bind(SOCKET_PATH)
    os.chmod(SOCKET_PATH, 0o666)
    self.socket_server.listen(5)
    print(f"[*] Æyre Media Daemon listening on socket: {SOCKET_PATH}")

    while self.running:
        try:
            conn, _ = self.socket_server.accept()
            threading.Thread(target=self.handle_client, args=(conn,), daemon=True).start()
        except Exception:
            break

def handle_client(self, conn: socket.socket):
    with conn:

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while self.running:
    data = conn.recv(4096)
    if not data:
        break
    try:
        req = json.loads(data.decode("utf-8"))
        cmd = req.get("command")
        if cmd == "SPEAK":
            text = req.get("text", "")
            pcm_bytes = self.synthesize_speech(text)
            resp = {"status": "SUCCESS", "bytes_count": len(pcm_bytes)}
            conn.sendall(json.dumps(resp).encode("utf-8"))
        elif cmd == "STATUS":
            resp = {"status": "ACTIVE", "vad_state": "LISTENING"}
            conn.sendall(json.dumps(resp).encode("utf-8"))
        else:
            conn.sendall(json.dumps({"status": "UNKNOWN_COMMAND"}).encode("utf-8"))
    except Exception as e:
        conn.sendall(json.dumps({"status": "ERROR", "message": str(e)}).encode("utf-8"))

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def start(self):
    self.running = True
    self.init_models()
    ipc_thread = threading.Thread(target=self.start_ipc_server, daemon=True)
    ipc_thread.start()
    print("[✓] Æyre Voice Stream Daemon active and listening.")
    try:
        while self.running:
            time.sleep(1)
    except KeyboardInterrupt:
        self.stop()

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def stop(self):
    print("\n[*] Stopping Æyre Voice Stream Daemon...")
    self.running = False
    if self.socket_server:
        self.socket_server.close()
    if os.path.exists(SOCKET_PATH):
        os.remove(SOCKET_PATH)
    print("[✓] Audio streams and sockets torn down.")

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if __name__ == "__main__":
    daemon = VoiceStreamDaemon()

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```
daemon.start()
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